Web Security Assessment

Target: testphp.vulnweb.com

Scope of Assessment:

- **Port Scanning**
- **Service Enumeration**
- **HTTP Header Analysis**
- **Directory Enumeration**

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Introduction

This assessment was conducted to identify open ports, running services, and potential security weaknesses on the target web application using structured reconnaissance and enumeration techniques. The objective was to evaluate exposed technologies and assess potential security risks.

This assessment was performed in a controlled lab environment for educational purposes.

Reconnaissance & Service Enumeration

Nmap Scan Findings:

- Open Port: 80/tcp
- Service: HTTP
- Server: nginx 1.19.0
- No additional open ports were identified during the scan.

HTTP Header Findings:

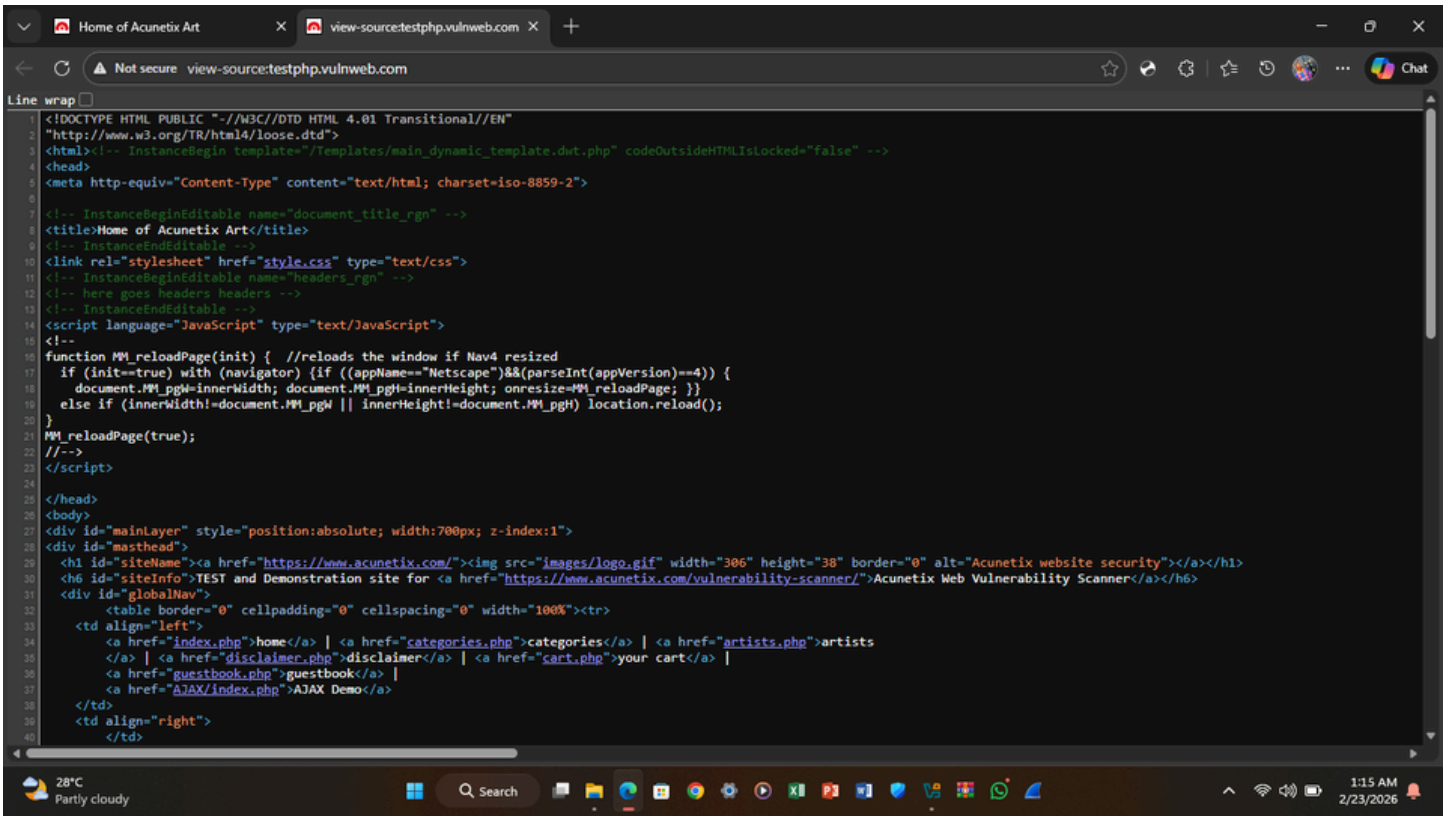
- Server: nginx/1.19.0
- X-Powered-By: PHP/5.6.40

The scan confirmed that the web server is publicly accessible over HTTP and is running specific software versions that are exposed through service detection and HTTP headers.

Web Application Analysis

- HTML 4.01 Transitional detected
- Character encoding: iso-8859-2
- Backend technology identified: PHP 5.6.40

The website utilizes older web technologies, including PHP 5.6.40, which is an end-of-life and unsupported version. Legacy technologies increase the likelihood of known vulnerabilities being present.



Directory Enumeration

Accessible directories identified during enumeration include:

- **/images/**
- **/pictures/**
- **/css/**
- **/js/**
- **/admin/**

While these directories are common in web applications, exposed administrative paths may present additional attack surfaces if not properly secured.

```
22:02
kali@kali: ~
Session  Actions  Edit  View  Help
└─$ dirb http://testphp.vulnweb.com

DIRB v2.22
By The Dark Raver

START_TIME: Sun Feb 22 19:51:16 2026
URL_BASE: http://testphp.vulnweb.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

---- Scanning URL: http://testphp.vulnweb.com/ ----
==> DIRECTORY: http://testphp.vulnweb.com/admin/
+ http://testphp.vulnweb.com/cgi-bin (CODE:403|SIZE:276)
+ http://testphp.vulnweb.com/cgi-bin/ (CODE:403|SIZE:276)
+ http://testphp.vulnweb.com/crossdomain.xml (CODE:200|SIZE:224)
==> DIRECTORY: http://testphp.vulnweb.com/CVS/
+ http://testphp.vulnweb.com/CVS/Entries (CODE:200|SIZE:1)
+ http://testphp.vulnweb.com/CVS/Repository (CODE:200|SIZE:8)
+ http://testphp.vulnweb.com/CVS/Root (CODE:200|SIZE:1)
+ http://testphp.vulnweb.com/favicon.ico (CODE:200|SIZE:894)
==> DIRECTORY: http://testphp.vulnweb.com/images/
+ http://testphp.vulnweb.com/index.php (CODE:200|SIZE:4958)
==> DIRECTORY: http://testphp.vulnweb.com/pictures/
==> DIRECTORY: http://testphp.vulnweb.com/secured/
==> DIRECTORY: http://testphp.vulnweb.com/vendor/

---- Entering directory: http://testphp.vulnweb.com/admin/ ----

---- Entering directory: http://testphp.vulnweb.com/CVS/ ----
+ http://testphp.vulnweb.com/CVS/Entries (CODE:200|SIZE:1)
+ http://testphp.vulnweb.com/CVS/Root (CODE:200|SIZE:1)

---- Entering directory: http://testphp.vulnweb.com/images/ ----

---- Entering directory: http://testphp.vulnweb.com/pictures/ ----
-> Testing: http://testphp.vulnweb.com/pictures/happening
```

Risk Assessment

Identified Risks:

- **Outdated PHP version (5.6.40)**
- **Server version disclosure (nginx 1.19.0)**
- **Exposure of web directories**

The use of unsupported software versions such as PHP 5.6 significantly increases the risk of exploitation, as publicly known vulnerabilities may be leveraged by attackers. Additionally, server version disclosure provides useful information that could aid reconnaissance efforts.

Recommendations

- Upgrade PHP to a currently supported and maintained version
- Update nginx to the latest stable release
- Disable server version disclosure in HTTP headers
- Implement regular patch management procedures
- Conduct periodic security assessments

Proper maintenance and timely updates significantly reduce the attack surface of web applications

Conclusion

The assessment identified exposed services and outdated technologies that increase the potential security risk of the web application. While no exploitation was performed, the presence of unsupported software versions and publicly disclosed server information highlights the importance of proactive security management.

Regular updates, secure configuration practices, and continuous monitoring are essential to maintaining a secure web environment.